

Accelerated Drug Discovery

On-line TLC/CMS technique using the Advion Interchim Scientific® expression® CMS and Plate Express™

This application explores the use of alternative chromatography technology for faster, more efficient analysis over traditional LC/MS runs.

Plate Express™ is a device that provides a simple, automated means of obtaining mass spectra directly from TLC plates, combined with the Advion Interchim Scientific® expression® CMS (Compact Mass Spectrometer) creating a technique known as TLC/CMS. Using this technique chemists can quickly and confidently identify products even in complex mixtures without additional sample preparation.

By decoupling the separation from the analysis, medicinal chemists can quickly analyse a developed TLC plate for answers in seconds. Not only less time-consuming, but the process is also much lower cost, with significantly less solvent consumption.

Keywords: Chromatography, LC/MS, TLC/CMS, Advion Interchim Scientific®

Introduction to TLC/CMS vs LC/MS

TLC/CMS for the organic synthetic and medical laboratories

Thin layer chromatography (TLC) is used in many organic synthetic and medical laboratories because it is a simple, cost-effective technique that provides chemists with critical information about their synthetic reactions. Structural characterization of the analytes by TLC is not possible by optical methods such as UV or ELSD. Typically, characterization is performed by GC/MS or LC/MS using sample preparation techniques which involves scraping the TLC spot of interest,

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extraction using suitable solvents, concentration, and then reconstitution in MS appropriate solvents. Here, an on-line TLC/CMS technique is presented using the Advion Interchim Scientific® expression® CMS and Plate Express™ to provide compound structural information without sample preparation after TLC separation.

A Suzuki reaction for the synthesis of 4-aminobiphenyl will be demonstrated. The online TLC/CMS technique provides rapid and accurate determination of reaction mixture components without the need for off-line TLC sample preparation procedures.

Method

Suzuki Reaction Setup

Reactants A and B were mixed at equimolar amounts in a round-bottom reaction flask and stirred at room temperature. 2 mL aliquots were transferred from the flask and spotted onto a Merck TLC Silica gel 60 F254 plate (10 x 10cm). Chemicals were purchased from Sigma- Aldrich with a purity greater than 99%. MS solvent was LC/MS grade.

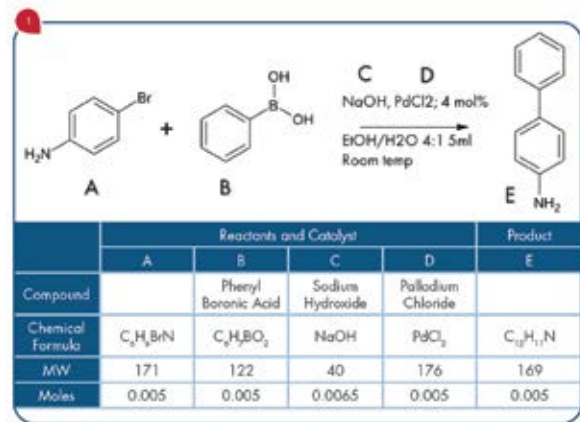


Figure 1: Chemical structure and information of reactants and catalysts for the Suzuki reaction

TLC Plate Preparation

Merck TLC Silica gel 60 F254 on Aluminium, 20 x 20 cm – cut to 10 x 10 cm. TLC plates were baked at 100 °C for 10 minutes to displace moisture. A 2 uL aliquot of the reaction mixture was spotted onto the TLC plate at the

position of 1 cm away from bottom. The TLC plate was then baked at 80 °C for 5 min. 30 mL Benzene was added to a TLC plate developing chamber at the level of 0.5 cm. The developing chamber was sealed with a glass lid for 30 min. The separation was stopped when the development solvent front reached a position which was 1 cm away from the top of the TLC plate.

The analytes (reactant and product) on the developed plate were observed under UV at 254 nm (Figure 2). The R_f value of the product (4-aminobiphenyl) was 0.1; the R_f of the reactant (4-bromoaniline) was 0.15.



Figure 2: The developed TLC plate with Suzuki reaction mixture.

TLC/CMS Method

The TLC/CMS analysis of the Suzuki reaction mixture at different reaction times was performed on the Advion Interchim Scientific® expression® CMS and Plate Express™. A solvent composed of 0.1% formic in methanol was used for the elution of the analytes from the TLC plate. The eluted analytes were directed to the CMS for acquisition of the corresponding mass spectra for the reactants and products.

Results

The mass spectra of the reactant and product are shown in Figure 3. The [M+H]⁺ for 4-bromoaniline was observed at m/z 171.9 (⁷⁹Br) and 173.9 (⁸¹Br). Methanol adducts were also detected at m/z 204.0 and 206.0 (Figure 4A). The [M+H]⁺ for 4-aminobiphenyl was observed at m/z 170.1 with a methanol adduct at m/z 202.1 (Notice the absence of the bromine doublet peaks in the non-brominated product of p-aminobiphenyl, Figure 1B).

The extracted ion current of the product ion at m/z 170.1 (protonated 4-aminobiphenyl) was monitored over different reaction times from 0 to 180 minutes (Figure 3A). The mass spectrum of the product ion at 120 min is

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shown in Figure 4B. The extracted ion current (XIC) of the reactant ion at the m/z 171.9 (protonated 4-bromaniline(^{79}Br)) is shown in Figure 4C.

The intensity of the m/z 171.9 decreased over the course of the reaction with respect to the product ion at the m/z 170.1. The reactant ion at the m/z 171.9 was no longer detected at 180 minutes, indicating the reaction was complete.

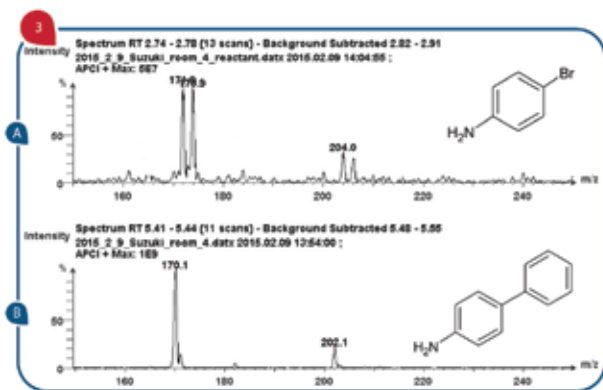


Figure 3: TLC/CMS analysis of reactant and product. (A) The mass spectrum of protonated 4-bromoaniline (reactant) at the m/z 171.9 and 173.9. (B) The mass spectrum of protonated 4-aminobiphenyl (product) at the m/z 170.1.

The extracted ion current of the product ion at m/z 170.1 (protonated 4-aminobiphenyl) was monitored over different reaction times from 0 to 180 minutes (Figure 4A). The mass spectrum of the product ion at 120 min is shown in Figure 4B.

The extracted ion current (XIC) of the reactant ion at the m/z 171.9 (protonated 4-bromaniline(^{79}Br)) is shown in Figure 4C. The intensity of the m/z 171.9 decreased over the course of the reaction with respect to the product ion at the m/z 170.1. The reactant ion at the m/z 171.9 was no longer detected at 180 minutes, indicating the reaction was complete.

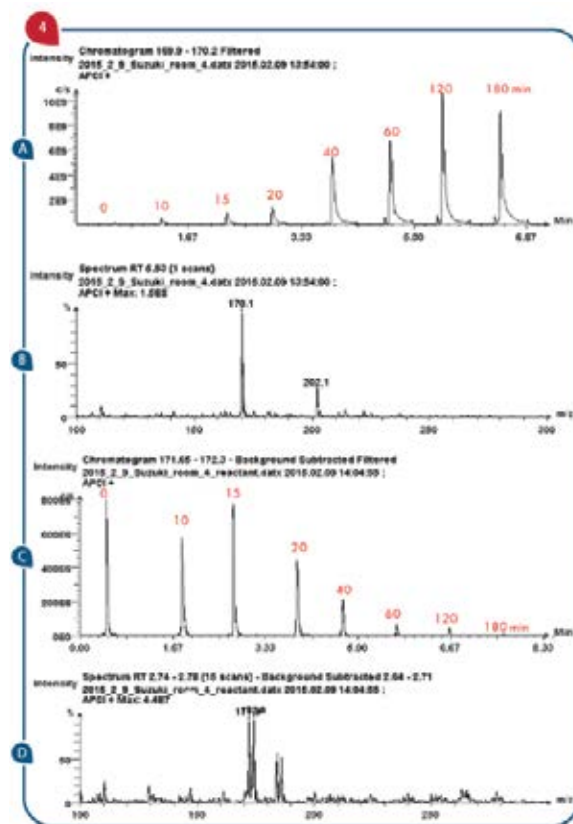


Figure 4: (A) XIC of product ion (protonated 4-aminobiphenyl), (B) mass spectrum of the product ion at reaction time at 120 min, (C) XIC of the reactant ion (protonated 4-bromoaniline), and (D) mass spectrum of the reactant ions at 15 minutes.

Conclusion

The Advion Interchim Scientific® expression® CMS coupled with the Advion Plate Express offers a simple and fast technique to monitor a Suzuki reaction for the synthesis of 4-aminobiphenyl.

The Advion Interchim Scientific® TLC/CMS system allows the synthetic chemist to monitor the reaction in real-time by evaluating the mass spectra for structural information (i.e., relative intensity of reactants vs. product) directly from the TLC plate.

The compact size allows it to fit into space-limited labs for direct access and immediate results for chemists requiring mass confirmation, reaction monitoring, quality control and purity analysis.

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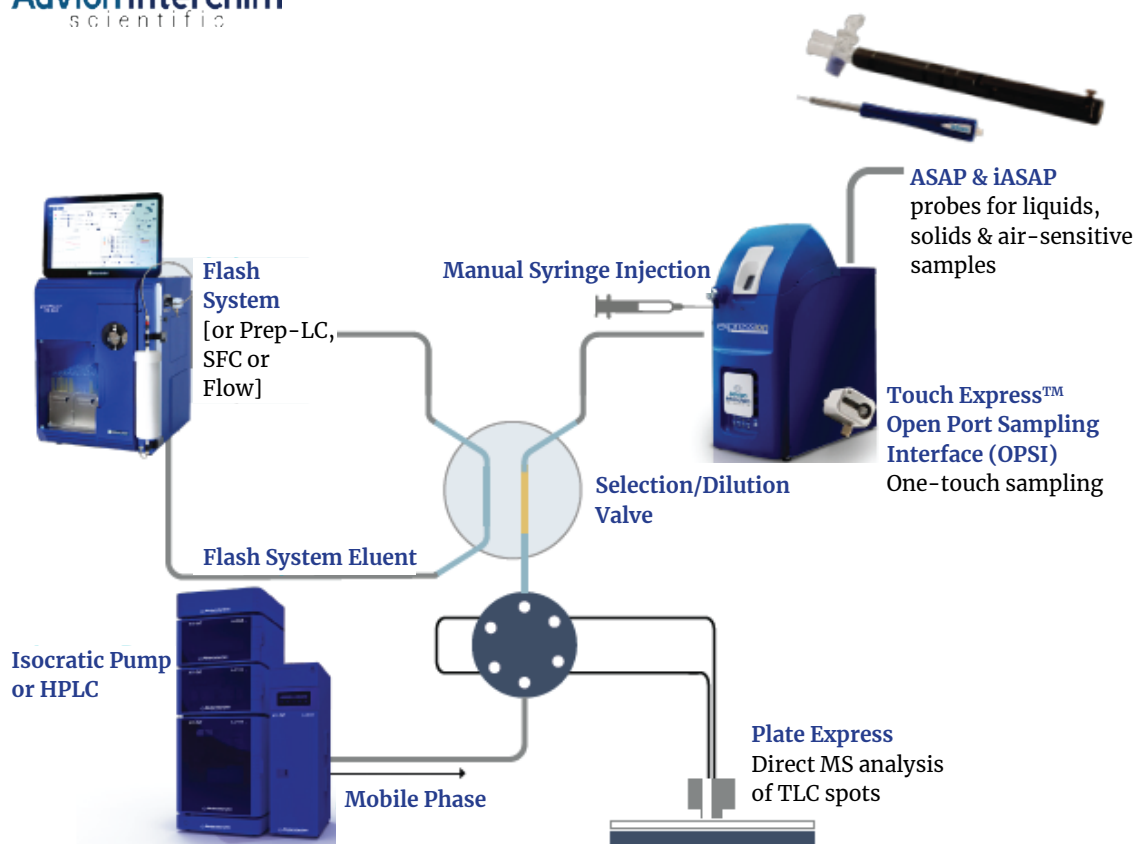
Multi-mode Mass Spectrometry Simultaneous integration with several techniques

Placed at the center of the chemist's workflow, the expression® CMS can provide critical information for many processes, reaction monitoring, separation, purification, and impurity determination.

It can be integrated with several techniques at once and the chemist can move effortlessly between these techniques in seconds without the need to re-plumb or re-configure the system, as below:

- Manual syringe injection
- Direct analysis probe (ASAP and iASAP probes)
- TLC/CMS with the Plate Express
- Flash chromatography, Prep-LC or SFC
- VOCs in air, breath and headspace (vAPCI)
- (U)HPLC


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Combined flash chromatography, TLC, (U)HPLC and manual injection interface. No re-plumbing necessary.